

One-to-one and onto

Definition

We say that a transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is

- **one-to-one** if whenever $u \neq v \in \mathbb{R}^n$ then $T(u) \neq T(v)$, and
- **onto** if the range of T is $\mathbb{R}^m$ i.e. for every $b \in \mathbb{R}^m$, there is $x \in \mathbb{R}^n$ such that $T(x) = b$.

Fact: If T is a linear transformation then the following are equivalent:

- for all $u \in \mathbb{R}^n$, if $T(u) = 0$ then $u = 0$.
- T is one-to-one.

Ex: $T\begin{pmatrix} x \\ y \end{pmatrix} = \underline{\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}} \begin{pmatrix} x \\ y \end{pmatrix} \leftarrow \text{matrix transformation.}$
 from $\mathbb{R}^2 \rightarrow \mathbb{R}^2$.

Is T 1-1?

Check: If $T\begin{pmatrix} x \\ y \end{pmatrix} = 0$ is it

$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0$ has only 0 as

a solution then T is 1-1.

$\text{rref} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Yes T is
1-1.

One-to-one and onto in terms of the standard matrix

Theorem

Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation, and A is the standard matrix for T . Then

- T is one-to-one iff the columns of A are linearly independent, and
- T is onto iff $\mathbb{R}^m$ if the span of the columns of A .

$T(u) = Au$ if the only solution to $Au = 0$ is $u = 0$. The columns of A are l.n. indep.

Ex: Continuing the previous example

$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Are these columns
lin. indep.?

Yes because they are not scalar mults.

Let $A = (u_1, u_2, \dots, u_n)$ where $u_1, \dots, u_n \in \mathbb{R}^n$

The range of T_A has the form - all vectors that look like $T_A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \underbrace{x_1 u_1 + \dots + x_n u_n}_{\text{lin. comb. of } u_1, \dots, u_n}$

These are the vectors in the span of $u_1, \dots, u_n$.

① the range of T_A is the span of its columns.

② T_A is onto iff $\mathbb{R}^n$ is the span of $u_1, \dots, u_n$.

Ex: $A = \begin{pmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \end{pmatrix}$. Is T_A 1-1?

No. Too many columns. If we solve $Au=0$

we get 2 pivots at most and so we have
free variables $\therefore$ non-trivial solutions!

Is T_A onto? Yes! Any two columns
span $\mathbb{R}^2$.

$$\text{Ex: } T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ 2x_1 - x_2 \\ 3x_2 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

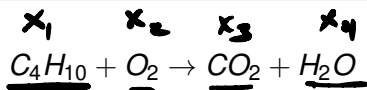
$$= \begin{pmatrix} 1 & 1 \\ 2 & -1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$\nearrow$
 A

1) Is T_A 1-1? Two columns are lin. indep.

2) Is T_A onto? No! Only 2 columns - need at least 3 to get all of $\mathbb{R}^3$.

An application: Balancing chemical equations



Butane ignited yields carbon dioxide and water

$$C: 4x_1 = x_3$$

$$H: 10x_1 = 2x_4$$

$$O: 2x_2 = 2x_3 + x_4$$

$$\begin{pmatrix} 4 & 0 & -1 & 0 \\ 10 & 0 & 0 & -2 \\ 0 & 2 & -2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$x_1 = \frac{1}{5}x_4 \quad x_2 = \frac{13}{10}x_4 \quad x_3 = \frac{4}{5}x_4.$$

A solution

is

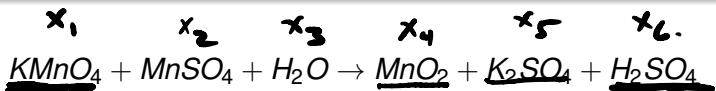
$$x_1 = 2$$

$$x_2 = 13$$

$$x_3 = 8$$

$$x_4 = 10$$

An application: Balancing chemical equations



Potassium permanganate and Manganese sulfate mixed in water yields manganese dioxide, potassium sulfate and sulphuric acid

$$\text{K} : x_1 = 2x_5$$

$$\text{Mn} : x_1 + x_2 = x_4$$

$$\text{O} : 4x_1 + 4x_2 + x_3 = 2x_4 + 4x_5 + 4x_6$$

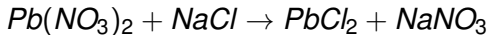
$$\text{S} : x_2 = x_5 + x_6$$

$$\text{H} : 2x_3 = 2x_6$$

A solution is $(2, 3, 2, 5, 2, 2)$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & -2 & 0 \\ 1 & 1 & 0 & -1 & 0 & 0 \\ 4 & 4 & 1 & -2 & -4 & -4 \\ 0 & 1 & 0 & 0 & -1 & -1 \\ 0 & 0 & 2 & 0 & 0 & -2 \end{pmatrix}$$

An application: Balancing chemical equations



Lead nitrate and sodium chloride mixed yields lead chloride and sodium nitrate